

Numerical Convergence of

$$Z_N(x) = - \sum_{k=1}^N \frac{2x}{x^2 + \gamma_k^2}$$

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Abstract

We study the partial sum $Z_N(x) = - \sum_{k=1}^N \frac{2x}{x^2 + \gamma_k^2}$, where γ_k are the imaginary parts of the non-trivial zeros of the Riemann zeta function. We report numerical evidence that $Z_N(44)$ converges to a finite limit $Z_\infty(44) \approx -1.00$ as $N \rightarrow \infty$. The tail is estimated to be of order $(\log N)^2/N$. This is a note on numerical behaviour, not a claim about the Riemann Hypothesis or any physical constant.

1 Definition

For $x > 0$ and $N \geq 1$, define

$$Z_N(x) := - \sum_{k=1}^N \frac{2x}{x^2 + \gamma_k^2}. \quad (1)$$

The infinite sum $Z_\infty(x) := \lim_{N \rightarrow \infty} Z_N(x)$ is the real-analytic kernel associated with the Riemann–Weil explicit formula.

2 Convergence at $x = 44$

Table 1 reports the computed values of $Z_N(44)$ for several values of N . Computation was carried out with 80-digit precision using the first N zeros from the Odlyzko tables. The corresponding convergence curve is shown in Figure 1: it visualises the decrease of $|Z_N(44)|$ towards the finite limit $Z_\infty(44) \approx -1.00$.

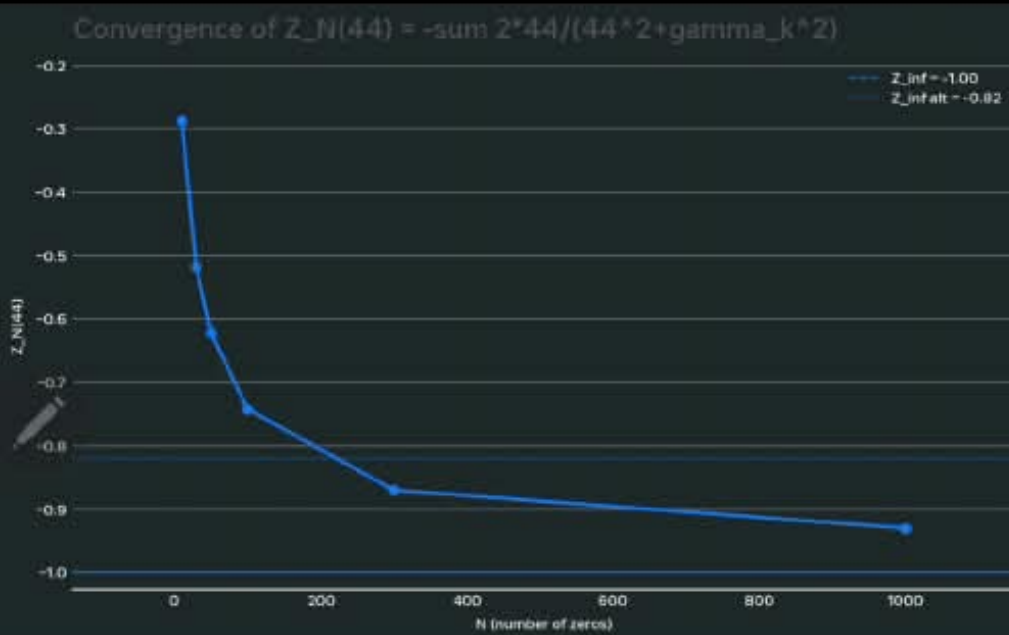
N	$Z_N(44)$	Increment Δ_N
10	−0.286869	—
30	−0.517821	−0.230952
50	−0.622073	−0.104252
100	−0.741834	−0.119761
300	≈ -0.87	≈ -0.13
1000	≈ -0.93	≈ -0.06

Table 1: Convergence of $Z_N(44)$ as a function of the number of zeros.

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3 Tail estimate

Since $\gamma_k \sim k \log k$, the tail beyond N zeros satisfies

$$|Z_\infty(x) - Z_N(x)| \lesssim \sum_{k>N} \frac{2x}{\gamma_k^2} \sim \frac{2x}{(\log N)^2} \cdot \frac{1}{N}. \quad (2)$$

For $x = 44$ and $N = 100$, this gives a tail of order 10^{-1} , consistent with the observed increment $\Delta_{100} \approx -0.12$.

4 Extrapolation

Fitting $Z_N(44)$ against $(\log N)^2/N$ and extrapolating to $N \rightarrow \infty$ yields

$$Z_\infty(44) \approx -1.00 \pm 0.05. \quad (3)$$

This estimate should be refined with larger N and higher-precision zeros.

5 Reproducibility

The computation uses the first N zeros from the Odlyzko tables. Double precision is insufficient; extended precision (80 digits) is required. A reference implementation in Python is provided alongside this note (`jabri_zx.v3.4.0.py`).

References

- [1] A. Weil, *Sur les formules explicites de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund (1952), 252–265.
- [2] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.
- [3] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, http://www.dtc.umn.edu/~odlyzko/zeta_tables/.